OPENING THE BOXES · SHADE AND SPONGE

How one replaced street tree connects to data-driven decisions, and back

reasoning step · a warning it must respect · what the research gives back

BIG TOPIC

Data-driven decision making

Cities set targets with data, but each tree is still chosen by rule of thumb

Which data?

Who decides?

URBAN CLIMATE DATA

Too hot to walk, water on the street: now mapped as UTCI and mm of runoff

ICGC LiDAR 2021; PDISBA storms; Ladybug UTCI

WHY THE PLANES MUST GO (CITY PLAN)

- **Pests:** one species, one pest; "vulnerability due to monoculture" (guide p. 17)
- **Plane-specific:** lace bug, powdery mildew, canker; sudden leaf drop in drought (guide pp. 37–38)
- **Biodiversity:** no species above 15% by 2037 (plan, target 3)
- **Climate:** adapted species 30% → 40% (plan, target 2)

THE CITY'S RULES

No species above 15%, tree size set by sidewalk width

Tree master plan 2017; street-tree guide 2011

what is missing?

what is left open?

INSIGHT

The data exists, but nobody links it to the species planted in each pit

why now?

PROBLEM

The plane is the only species over the city's 15% cap. Each replacement changes shade and runoff, and nobody measures it

Tree master plan 2017, target 3; street-tree guide 2011, p. 54

INSIGHT

Rules fix how much of each species, not which species where

who models both?

use measured crowns

use the city's rules

can a tool choose?

must follow the rules

HEAT + RUNOFF WORKFLOWS

Both services, fixed scenarios

Limit: no species choice

Mannucci 2025; Nyelele & Kroll 2021

IDEA: COMBINE BOTH SIDES

Replace each plane with the rule-compliant species that loses the least shade and rain interception

SPECIES TOOLS

Species recommenders and heat optimisers

Limit: never heat and runoff together

SylvCiT 2026; Shaamala 2025; Peng 2026

but not one service

gives back: a number for each choice the city cannot see now

the city's rule-following replacement is the baseline to beat

LITERATURE FUNNEL

Tools that place or assess urban trees

My prior-art matrix: 18 tools

who did it?

does it work?

which cover both?

7 heat and runoff

which pick species?

1 also picks species

which work per tree?

0 per tree, inside city rules

gives back: honest gains, small and measured

GAP · METHODOLOGICAL

No method picks species per tree position on heat and runoff together, inside a city's replacement rules

tests the cost of the 15% cap

so I ask

RESEARCH QUESTION

Can a trait-based layout, inside Barcelona's rules, keep more cooling and rain interception than the city's own replacement?

where else?

WHERE ELSE?

- 33 Barcelona neighbourhoods
- Transfer: Mataró, Florence
- Any city with a species cap
- More services: carbon, biodiversity

H1
Better on heat and runoff?

H2
Do the heat-best and runoff-best differ?

Limit
Gain shrinks as storms grow